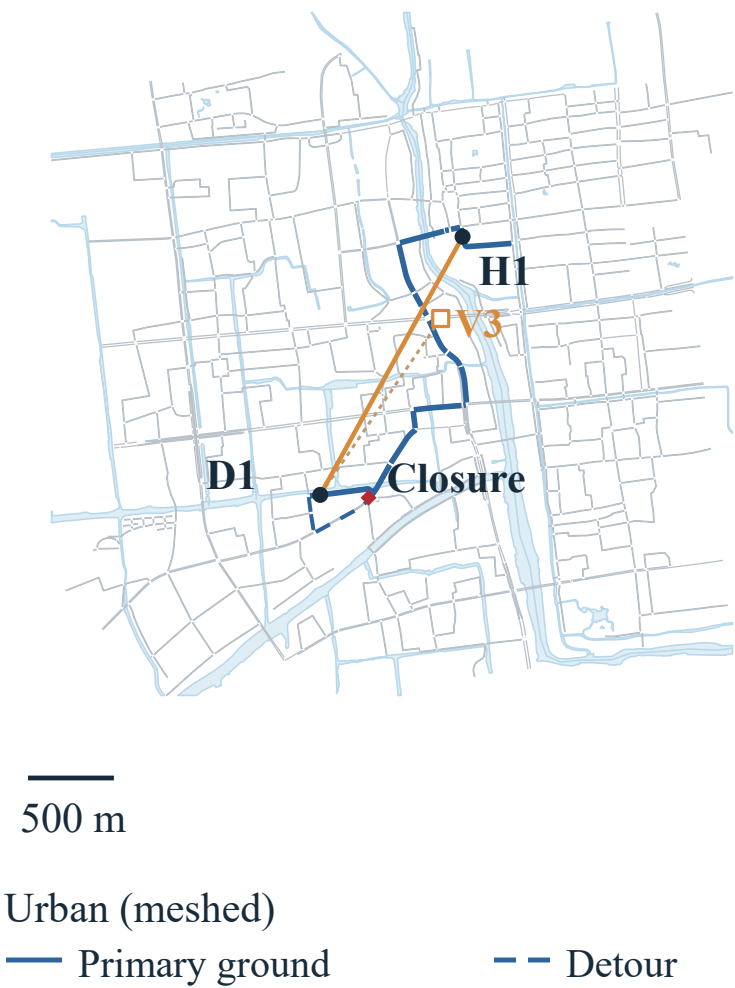
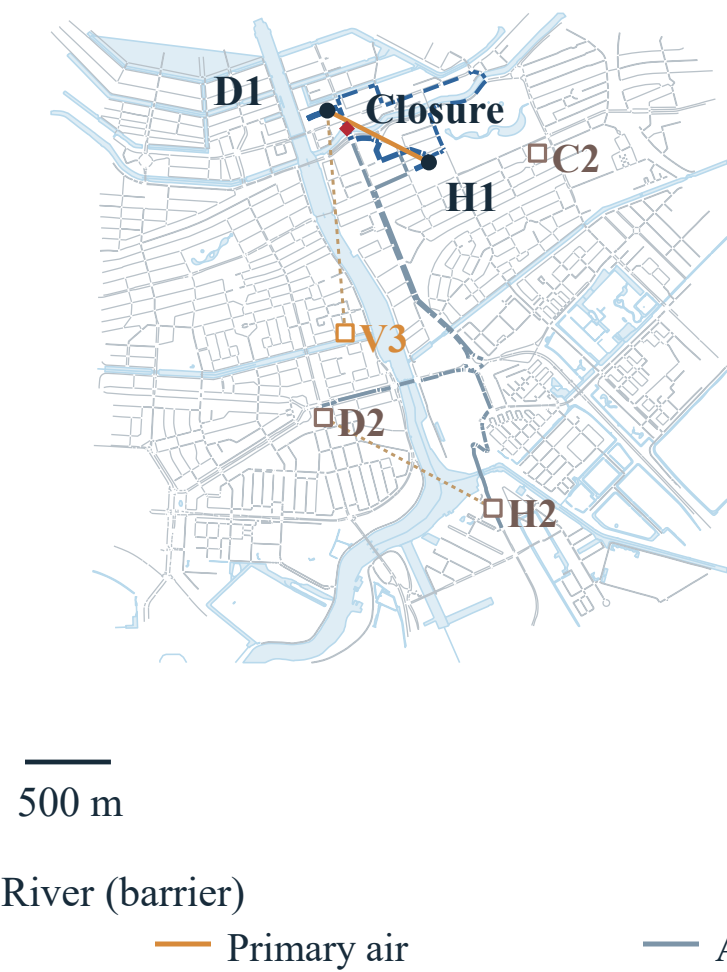


Contrasting network morphology

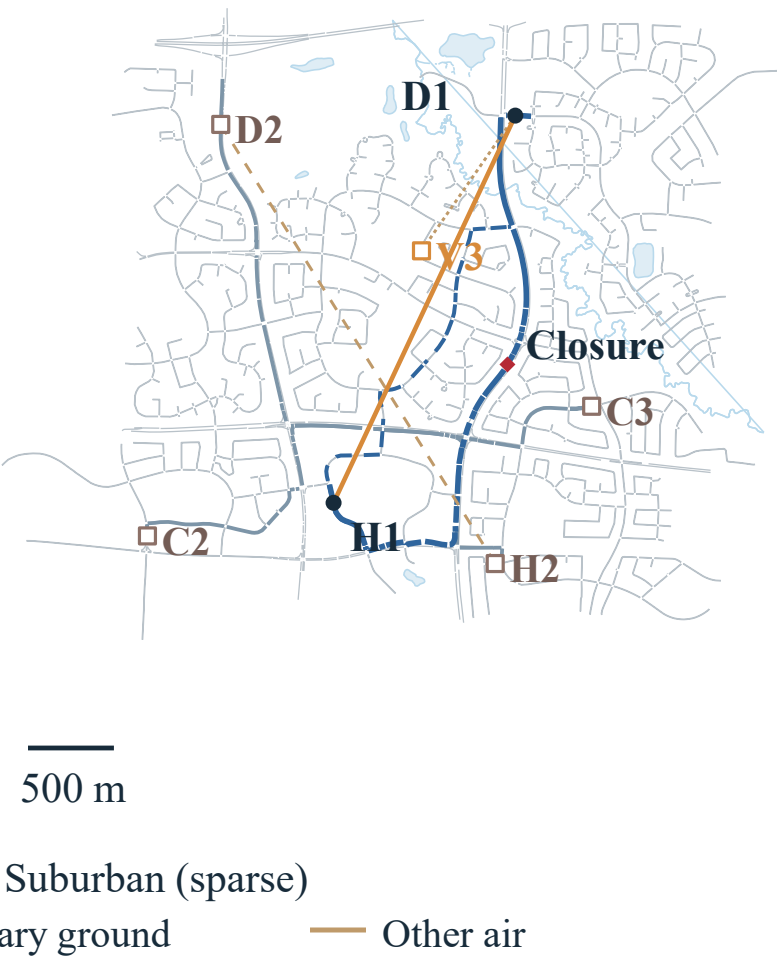
(a) Site A: Suzhou



(b) Site B: Amsterdam



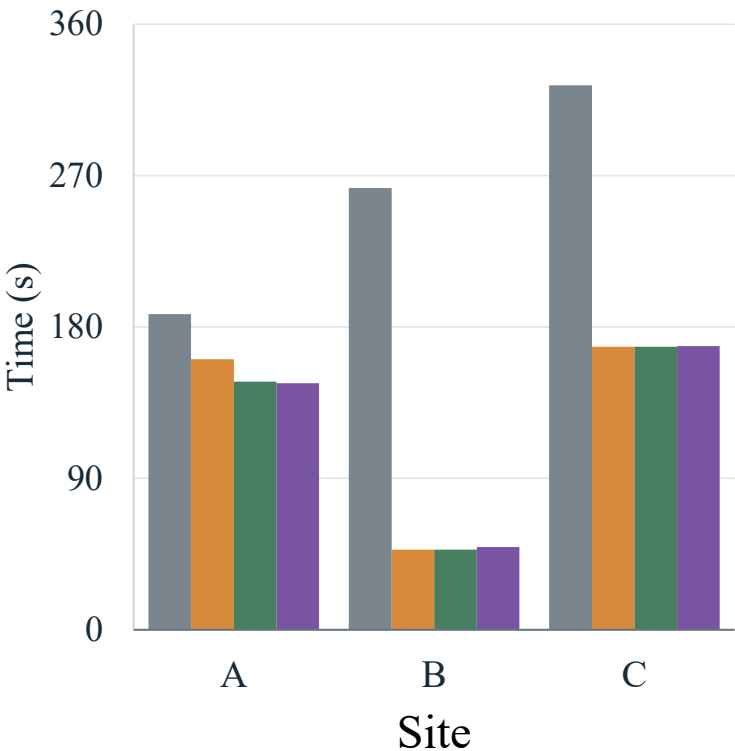
(c) Site C: Edmonton



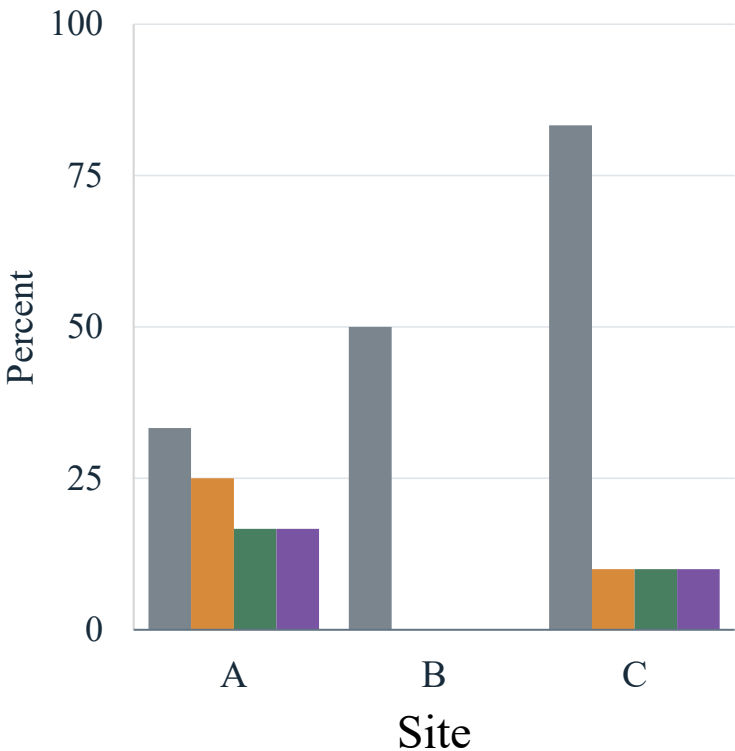
E1: selective air support and incumbent-service cost

■ B0 Ground ■ B1 Air-first ■ B2 Heuristic ■ B4b LLM

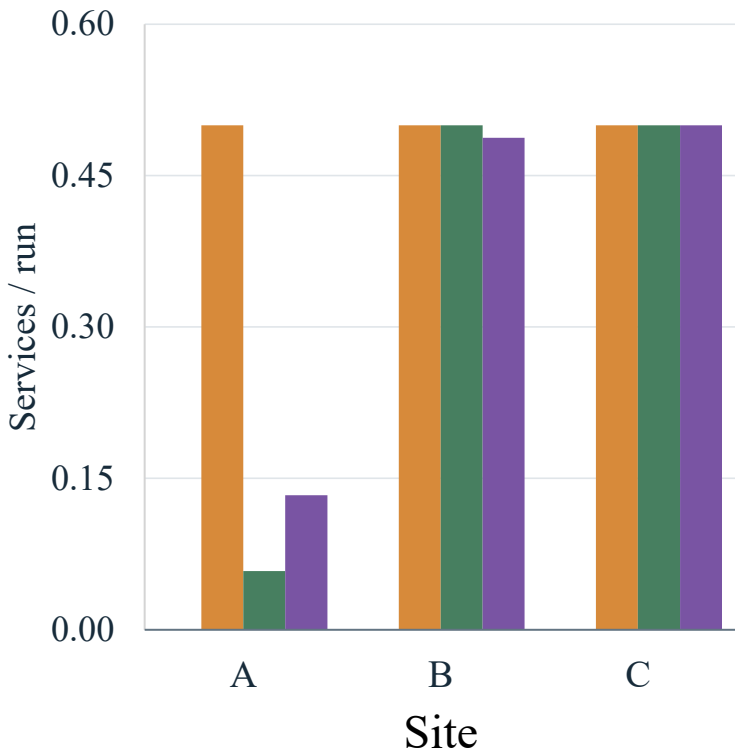
(a) Emergency completion



(b) Deadline violations

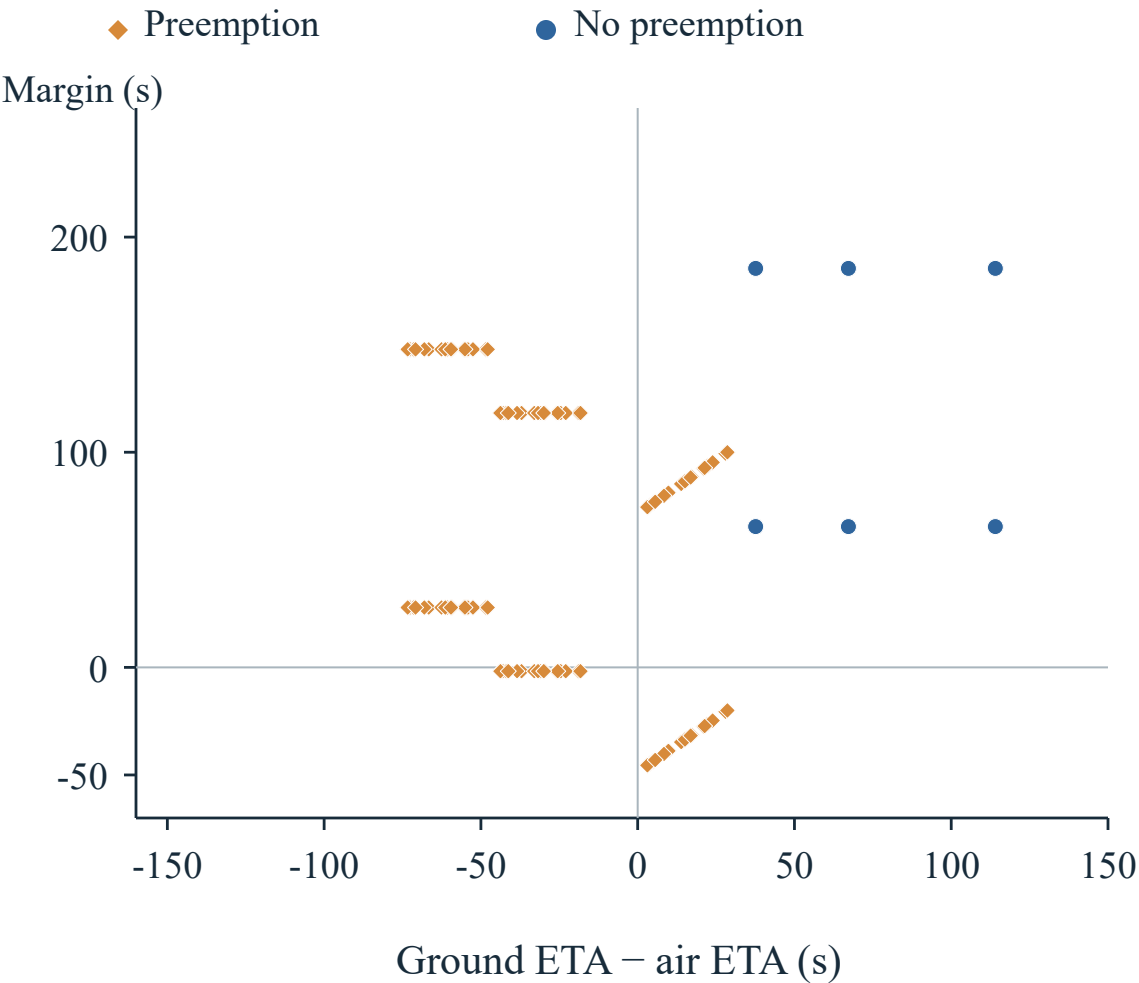


(c) Service damage

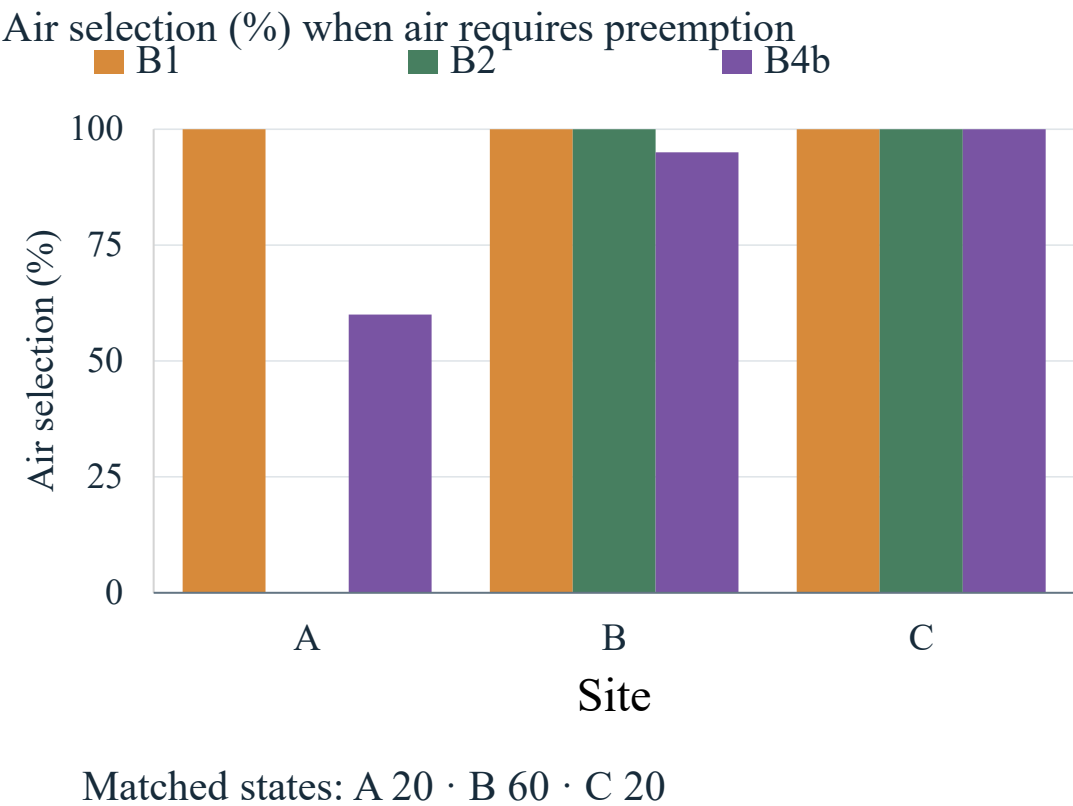


Supervisory decision regimes

(a) Candidate geometry at Site A



(b) Faster air vs timely ground



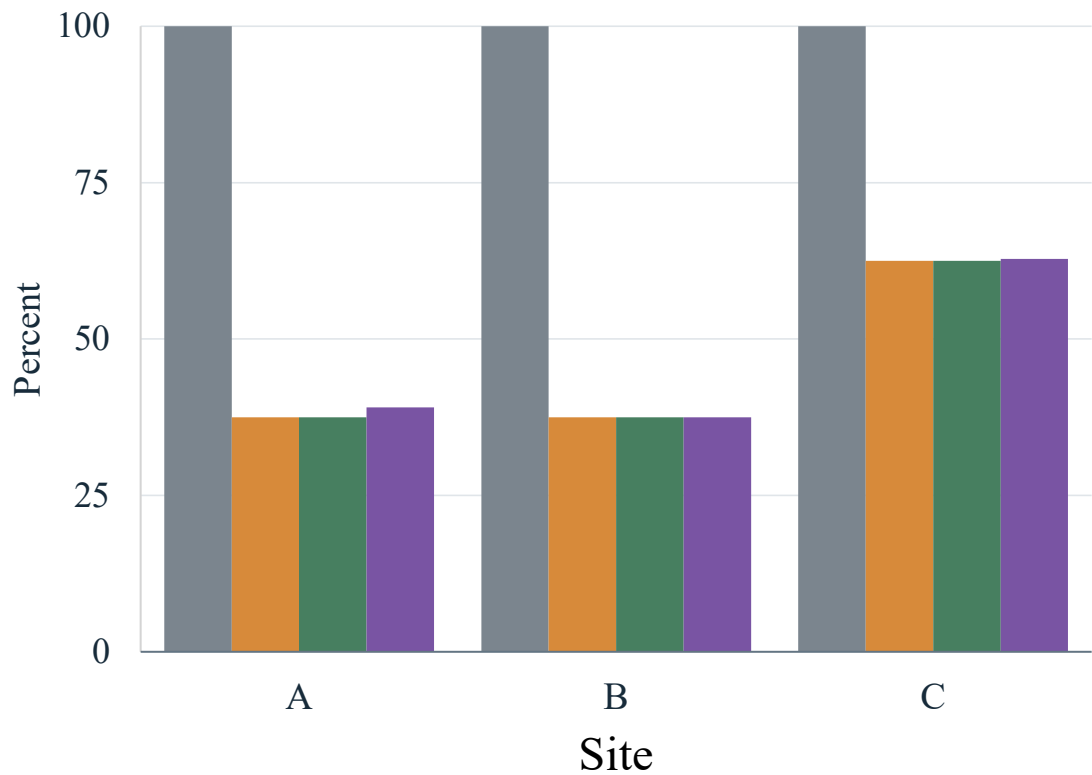
(c) No candidate predicted timely

Observed late: A 40/40 · C 24/27 per policy

E2: restoring a path and meeting the deadline

B0 Ground B1 Air-first B2 Heuristic B4b LLM

(a) Deadline violations in all runs



(b) Recovered / affected critical chains

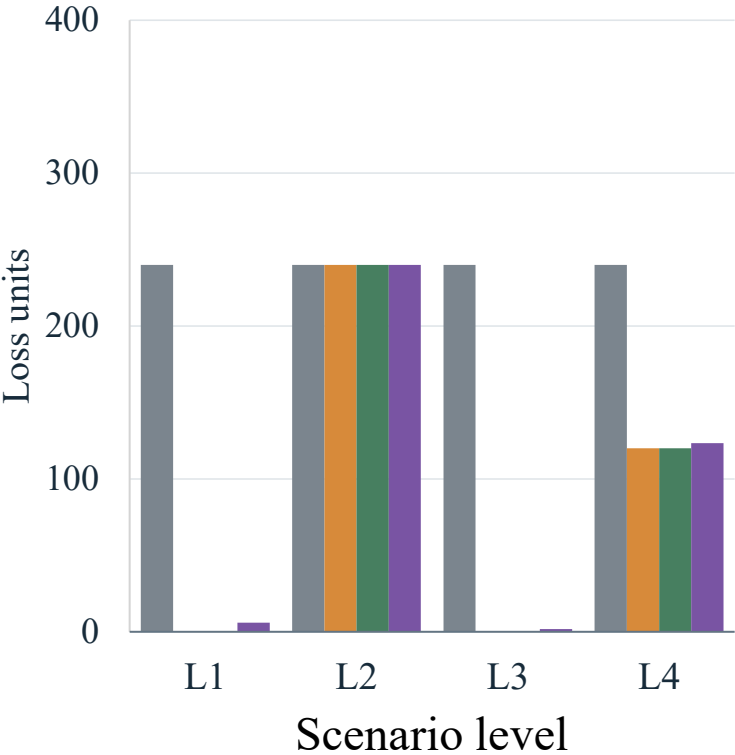
Site	Rule (B1)	Heuristic (B2)	LLM (B4b)
A	200/200	200/200	191/191
B	200/200	200/200	199/199
C	200/200	200/200	201/201

Every affected coordinated run restores a path
B0 recovery: not applicable

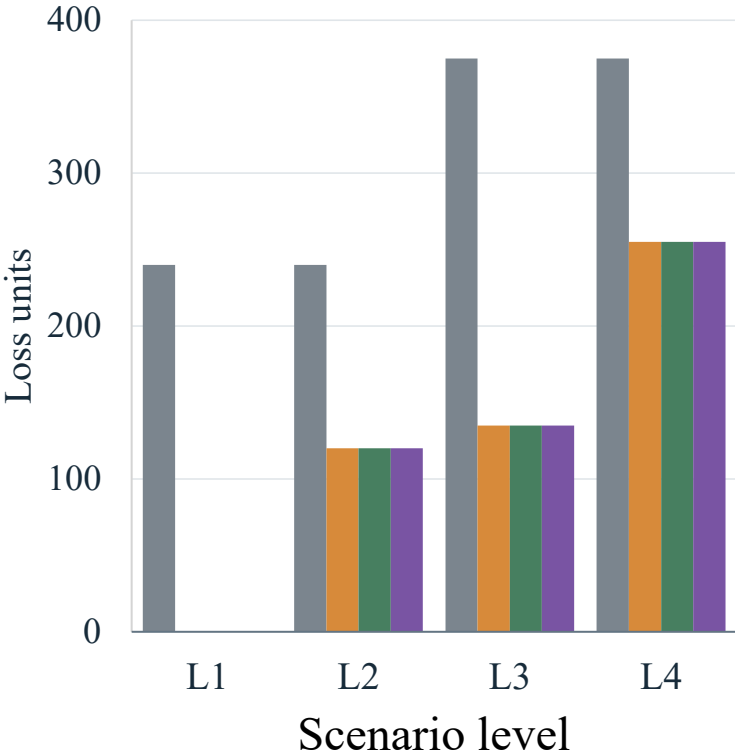
E3: priority-weighted service loss across network contexts

B0 Ground B1 Air-first B2 Heuristic B4b LLM

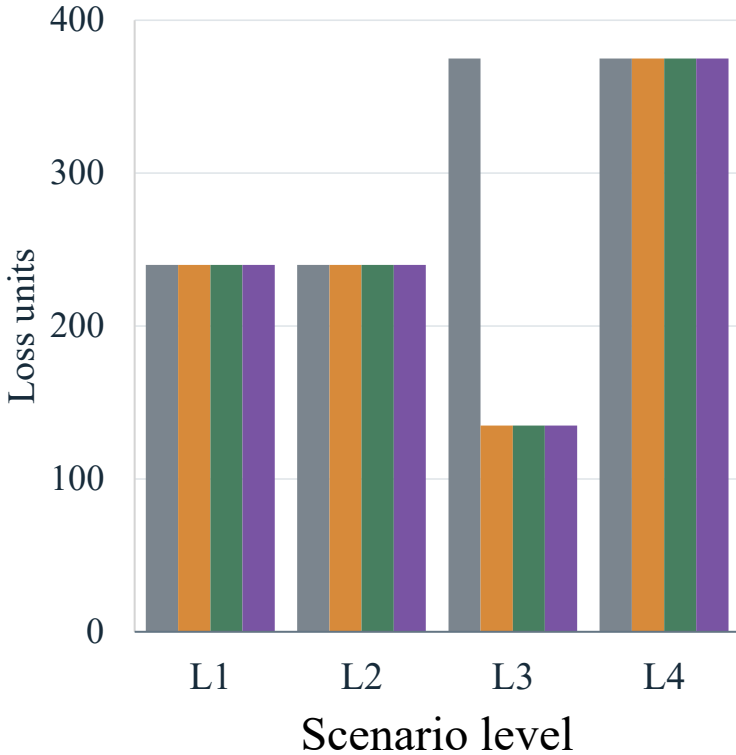
(a) Site A



(b) Site B



(c) Site C



E4: observation, execution and input burden

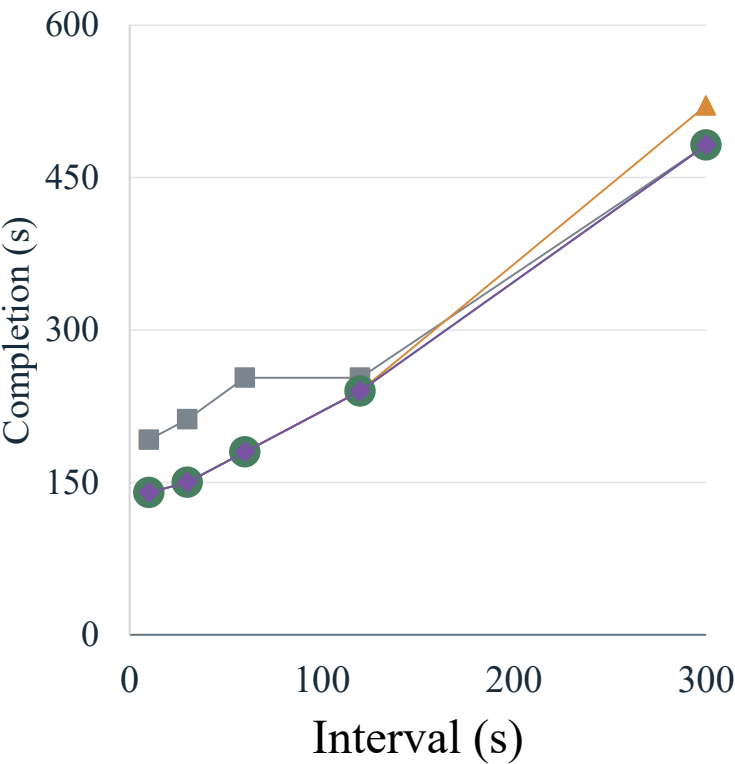
■ B0 Ground

■ B1 Air-first

■ B2 Heuristic

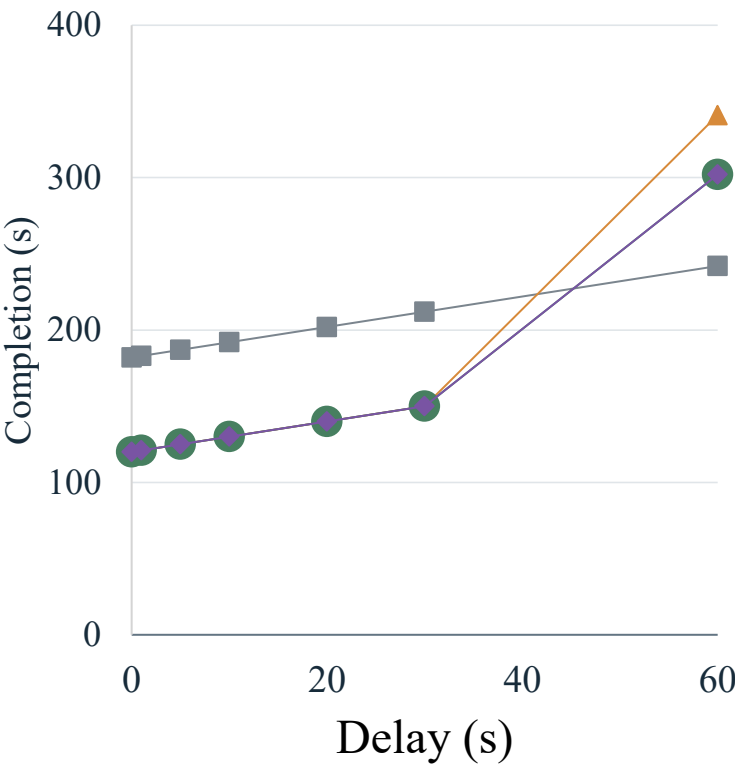
■ B4b LLM

(a) Observation interval



B2 = B4b (all arms)

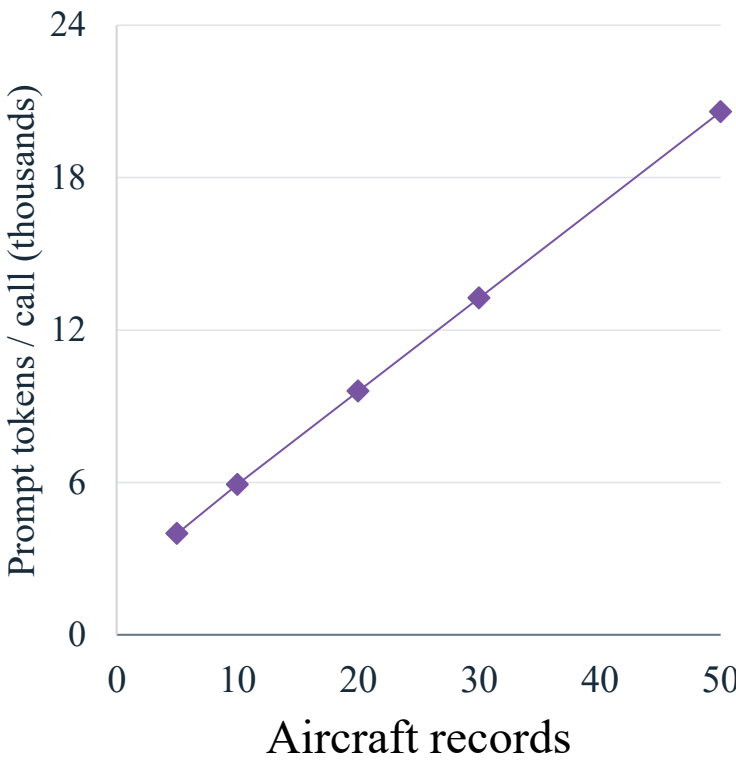
(b) Execution delay



B1 = B2 = B4b (0–30 s)

B2 = B4b (60 s)

(c) LLM input burden



Execution waiting: repeated command or changed intervention

